

2. F) $q=10$; $S = X^a Z^b$; $S^5 = I$
eigenvalues of $S = e^{2\pi i j/5}$ for $j = 0, \dots, 4$.

i. What are a & b ?

error span $\{I, X, X^{-1}, Z, Z^{-1}\}$

eigenvalues of S $\{1, e^{2\pi i/5}, e^{4\pi i/5}, e^{6\pi i/5}, e^{8\pi i/5}\}$

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$$Z|j\rangle = e^{2\pi i j/10} |j\rangle = e^{\pi i j/5} |j\rangle$$

$$X|j\rangle = |j+1 \bmod 10\rangle$$

$$XZ = ZX e^{-2\pi i/10} = ZX e^{-i\pi/5}$$

$$\text{errors} = \{I, X, X^{-1}, Z, Z^{-1}\}$$

$$\text{stabilizer} = X^a Z^b = S$$

$$SI = IS$$

$$SX = X^a Z^b X = \left(e^{\pi i/5}\right)^b \cdot X \underbrace{X^a Z^b}_S$$

$$SX = \left(e^{\pi i/5}\right)^b \cdot X \cdot S$$

$$SX^{-1} = X^a Z^b X^{-1}$$

$$\left[\text{Now, } XZ = ZX e^{-\pi i/5} \right.$$

$$X^{-1}XZ = X^{-1}ZX e^{-\pi i/5}$$

$$ZX^{-1} = X^{-1}ZX X^{-1} \cdot e^{-\pi i/5}$$

$$ZX^{-1} = X^{-1}Z \cdot e^{-\pi i/5} \left. \right]$$

$$\begin{aligned}
 \text{So, } SX^{-1} &= X^a Z^b \cdot X^{-1} \\
 &= X^a \cdot X^{-1} \cdot \left(e^{-\pi i/5}\right)^b \cdot Z^b \\
 &= \left(e^{-\pi i/5}\right)^b \cdot X^{-1} \cdot \underbrace{X^a \cdot Z^b}_S \\
 &= \left(e^{-\pi i/5}\right)^b \cdot X^{-1} \cdot S
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } SZ &= X^a Z^b \cdot Z \\
 &= X^a \cdot Z \cdot Z^b \\
 &= \left(e^{-\pi i/5}\right)^a \cdot Z \cdot \underbrace{X^a Z^b}_S \\
 &= \left(e^{-\pi i/5}\right)^a \cdot ZS
 \end{aligned}$$

$$\begin{aligned}
 \& \quad SZ^{-1} &= X^a Z^b \cdot Z^{-1} \\
 &= X^a \cdot Z^{-1} \cdot Z^b
 \end{aligned}$$

$$\begin{aligned}
 [\text{Now, } XZ &= ZX e^{-\pi i/5} \\
 Z^{-1} X Z &= X e^{-\pi i/5}
 \end{aligned}$$

$$z^{-1}x = xz^{-1} e^{-\pi i/5}]$$

$$\begin{aligned} \therefore S z^{-1} &= x^a z^{-1} \cdot z^b \\ &= \left(e^{\pi i/5} \right)^a \cdot z^{-1} \cdot \underbrace{x^a \cdot z^b}_S \\ &= \left(e^{\pi i/5} \right)^a \cdot z^{-1} \cdot S \end{aligned}$$

Now, given eigen-values

$$= \left[1, e^{2\pi i/5}, e^{4\pi i/5}, e^{6\pi i/5}, e^{8\pi i/5} \right]$$

Matching them with the values we have:

$$\left[1, \left(e^{\pi i/5} \right)^b, \left(e^{\pi i/5} \right)^{-b}, \left(e^{\pi i/5} \right)^{-a}, \left(e^{\pi i/5} \right)^a \right]$$

$$X^{10} = I ; X^{10-a} = X^{-a} ; Z^{10-b} = Z^{-b}$$

So, For $S = X^a Z^b$, we only care about a & b values mod 10.

$$\Rightarrow \begin{cases} b, 10-b, 10-a, a \end{cases}$$

$$\Rightarrow \begin{cases} 2, 8, 6, 4 \end{cases}$$

$$S = X^a Z^b = X^2 \cdot Z^4 \text{ or } X^2 \cdot Z^6 \text{ or } X^8 \cdot Z^4$$

$$\text{or } X^8 \cdot Z^6$$

How to decode :

- ① Any error in the span would send the state to one of the eigenspaces of the stabilizer.
- ② All these eigenspaces are orthogonal, so once we have identified what error sector the state is in, we can converge on the error that would have caused it.
- ③ Once we have identified the error we apply $\{X^9, X^{11}, Z^9, Z^{11}\}$ to correct the respective $\{X, X^{-1}, Z, Z^{-1}\}$.

logical operators:

$$X_L^2 P = Z_L^2 P = P$$

$$X_L Z_L P = -Z_L X_L P$$

$$[X_L, S] = 0$$

$$[Z_L, S] = 0$$

$$S = X^2 Z^4$$

Let's assume $X_L = X^K$

$$X^K \cdot X^2 \cdot Z^4$$

$$\Rightarrow X^2 \cdot X^K \cdot Z^4$$

$$\Rightarrow X^2 Z^4 \cdot X^K \cdot (e^{-\pi i/5})^{4K}$$

$$X^K \cdot S = (e^{-\pi i/5})^{4K} \cdot S \cdot X^K$$

This needs to be = 1 for commutativity
 $K=5$ value satisfies that.

So, $X_L = X^5$.

Now, let's assume $Z_L = Z^K$

$$Z^K \cdot X^2 Z^4 = (e^{\pi i/5})^{2 \cdot K} \cdot X^2 Z^4 \cdot Z^K$$

again $K=5$ makes the factor = 1.

$$\text{So, } Z_L = Z^5.$$

$$\therefore X_L = X^5 ; Z_L = Z^5$$

... QED